

Two-Stage Stochastic Stable Matching

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Introduction

The Stable Matching Problem (e.g., [6, 7, 10]):

- **Marriage Instance** $(A, B, \succ)$: A set of students A and a set of schools B and a set of orders $\succ$ of each side over the other.
- **Goal**: compute a **stable matching**, i.e. a matching with no blocking pairs (pairs that prefer each other to their assigned partners).
- Problem **studied in many two-stage settings** including when agents of the same type enter or leave between stages [5], when preferences change between stages [3, 2] and more [8, 9, 4]. Our paper is the first to study the problem in a two-stage stochastic setting.

Our Real World Motivation:

- In practice, after an **initial matching** is proposed.
- **A subset of agents might leave** (e.g. each year around 20% of students assigned to San Francisco public middle schools choose not to use their assigned seats [1]; schools might also have unforeseen budget cuts).
- **New agents might enter** (e.g. new students come to the city; schools have new seats).
- For the students that did not “commit” to their initial matching, a **second round** re-matching is needed.

Formal Definition

- **First stage instance** $I_1 = (A, B, \succ)$.
- **Second-stage distribution** $\mathcal{D}$ from which a subset S of agents present in the second-stage is sampled (this subsumes the case of arriving agents).
- **Costs** $c_1, c_2 : A \cup B \rightarrow \mathbb{R}$ measuring the quality of the first and second stage matchings, and cost

$$(M_1, M_2) \rightarrow d_S(M_1, M_2) = \lambda \sum_{a \in A \cap S} [R_a(M_2(a)) - R_a(M_1(a))]^+$$

measuring dissatisfaction of students present in both stages from being downgraded to less preferred schools between the first-stage matching M_1 and the second-stage matching M_2 . R_a is the rank function of a .

- **Objective**:

$$\min_{M_1 \in \mathcal{M}_{I_1}} c_1(M_1) + \mathbb{E}_{S \sim \mathcal{D}} \left[\min_{M_2 \in \mathcal{M}_{I_2}} c_2(M_2) + d_S(M_1, M_2) \right]. \quad (2\text{STO})$$

Here $I_1 = (A, B, \succ)$ is the first-stage instance, $I_2 = (A \cap S, B \cap S, \succ)$ is the second-stage instance, and $\mathcal{M}_I$ is the set of stable matchings of the marriage instance I .

Explicit Second-Stage Distribution

In the case when the distribution $\mathcal{D}$ is given explicitly by the list of all possible scenarios and their probabilities, we solve the problem exactly by reduction to minimum s-t cut on a directed graph.

Theorem 1. There exists an algorithm such that given any instance $\mathcal{I}$ of (2STO) where the second-stage distribution is given explicitly solves the the problem in time polynomial in the input size.

A sketch of our algorithm is as follows.

Algorithm

1. For every marriage instance I , we construct a capacitated digraph called **the cut graph** $G_{\text{cut}}(I)$ in which finite capacity s-t cuts are in a one-to-one correspondence with stable matchings of I .
2. In the union of the cut graphs $G_{\text{cut}}(I_1)$ and $\{G_{\text{cut}}(I_2^S)\}_S$, add arcs between the nodes of $G_{\text{cut}}(I_1)$ and $G_{\text{cut}}(I_2^S)$ for every scenario S such that the total capacity of arcs crossing the cut corresponding to matchings M_1, M_2^S is exactly $d_S(M_1, M_2^S)$.
3. Add arcs between the nodes of $G_{\text{cut}}(I_1)$ (resp. $G_{\text{cut}}(I_2^S)$ for every S) such that the total capacity of arcs crossing the cut corresponding to the matching M_1 (resp. M_2^S) is exactly $c_1(M_1)$ (resp. $c_2(M_2^S)$).
4. Solve the minimum s-t cut problem over the resulting graph to get an optimal solution of (2STO).

We run numerical experiments on instances where $\mathcal{D}$ is joint independent Bernoulli distributions over $A \cup B$. The stable matchings computed by our algorithm have relatively small cost compared to the optimal first-stage matching in hindsight.

Sampling Oracle Model

When the distribution $\mathcal{D}$ is given implicitly by a sampling oracle, we show that the problem becomes **NP-Hard** by a reduction to the #P-Hard problem of counting the number of vertex covers of a graph.

Theorem 2. Unless P=NP, there exists no algorithm that given any instance $\mathcal{I}$ of (2STO) where the second-stage distribution is specified implicitly by a sampling oracle, solves the problem in time and number of calls to the sampling oracle that is polynomial in the input size. This hardness result holds even if the cost parameters λ and $\{c_1(ab)\}_{ab}, \{c_2(ab)\}_{ab}$ are in $\{0, 1\}$.

On the positive side, we use the **sample average approximation method (SAA)**, to get a pseudopolynomial algorithm with arbitrary good additive approximation guarantee under the sampling oracle model. In particular, we show the following theorem,

Theorem 3. There exists an algorithm that, given an instance $\mathcal{I}$ of (2STO) where the second-stage distribution is specified implicitly by a sampling oracle, and two parameters $\epsilon > 0$ and $\alpha \in (0, 1)$, gives a first-stage stable matching solution M_1 such that,

$$\mathbb{P}(\text{value}(M_1) \leq \text{value}(M_1^*) + \epsilon) \geq 1 - \alpha.$$

Where M_1^* is an optimal solution. The algorithm runs in time polynomial in $\max_{ab} |c_2(ab)|, \lambda, 1/\epsilon$ and $\log(1/\alpha)$ and the rest of the input.

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